

Total antioxidant content of alternatives to refined sugar.

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BACKGROUND: Oxidative damage is implicated in the etiology of cancer, cardiovascular disease, and other degenerative disorders. Recent nutritional research has focused on the antioxidant potential of foods, while current dietary recommendations are to increase the intake of antioxidant-rich foods rather than supplement specific nutrients. Many alternatives to refined sugar are available, including raw cane sugar, plant saps/syrups (eg, maple syrup, agave nectar), molasses, honey, and fruit sugars (eg, date sugar). Unrefined sweeteners were hypothesized to contain higher levels of antioxidants, similar to the contrast between whole and refined grain products. **OBJECTIVE:** To compare the total antioxidant content of natural sweeteners as alternatives to refined sugar. **DESIGN:** The ferric-reducing ability of plasma (FRAP) assay was used to estimate total antioxidant capacity. Major brands of 12 types of sweeteners as well as refined white sugar and corn syrup were sampled from retail outlets in the United States. **RESULTS:** Substantial differences in total antioxidant content of different sweeteners were found. Refined sugar, corn syrup, and agave nectar contained minimal antioxidant activity (<0.01 mmol FRAP/100 g); raw cane sugar had a higher FRAP (0.1 mmol/100 g). Dark and blackstrap molasses had the highest FRAP (4.6 to 4.9 mmol/100 g), while maple syrup, brown sugar, and honey showed intermediate antioxidant capacity (0.2 to 0.7 mmol FRAP/100 g). Based on an average intake of 130 g/day refined sugars and the antioxidant activity measured in typical diets, substituting alternative sweeteners could increase antioxidant intake an average of 2.6 mmol/day, similar to the amount found in a serving of berries or nuts. **CONCLUSION:** Many readily available alternatives to refined sugar offer the potential benefit of antioxidant activity.